COMPLETED WORK & RESEARCH UNDERSTANDING

Day 1 — Problem Understanding

- Studiedthe problemstatementanddefined the system inputs and outputs
- Identified underwater imagery, object labels and annotations as key inputs
- Defined expected outputs as enhanced imagery, object detections and tracking analysis where sequence data is available
- Designed an end-to-end software pipeline
- Prepared technical questions and assumptions requiring clarification

Day 2 — Literature Review

- Reviewedunderwaterimagingand computer vision challenges
- Studied image enhancement approaches
- Reviewed deep learning-based object detection methods
- Studied transfer learning for object detection
- Reviewed tracking methods such as SORT, DeepSORT and ByteTrack
- Shortlisted candidate methods for the proposed system

KEY FINDINGS & PROPOSED RESEARCH DIRECTION

KEY FINDINGS

- Underwater imageryis affectedby colordistortion, scattering, low contrast and poor visibility
- Image enhancement can improve visual quality but may not always improve object detection performance
- Transfer learning is suitable for object detection when training data and computational resources are limited
- YOLO-based detectors are suitable candidates for efficient object detection
- Tracking can be investigated when video or sequential data is available

REVISED RESEARCH QUESTION

Howdoes underwater imageenhancement affect deep learning-based object detection performance, and can an effective pipeline provide reliable underwater visual surveillance?

PROPOSED EXPERIMENTAL COMPARISON

Raw images → object detection

vs.

Enhanced images → object detection

EVALUATION METRICS

Precision · Recall · mAP@50 · mAP@50–95

BLOCKERS & NEXT PLAN

CURRENT BLOCKERS

- Final dataset details are not yet confirmed
- Object classes and annotation format need verification
- Availability of video or sequence data needs confirmation
- Tracking requirements need clarification
- Final evaluation metrics need confirmation

MENTOR CLARIFICATIONS REQUIRED

- Confirm the target definition
- Confirm dataset structure and annotation format
- Confirm the object classes
- Confirm the train/validation/test split
- Confirm whether tracking is mandatory
- Confirm the required evaluation metrics

NEXT PLAN

1. Obtain and inspect the dataset
2. Perform exploratory data analysis
3. Validate annotations and dataset structure
4. Establish a baseline pipeline
5. Begin image enhancement and object detection experiments
6. Compare raw and enhanced image detection performance